

Signals & Systems

Vector Form :-

$$X = [X(0), X(1), \dots, X(N-1)]^T$$

Complex Sequences

$$Z(n) = a(n) + j b(n) = \text{Re}\{Z(n)\} + j \text{Im}\{Z(n)\}$$

Polar Form in terms of its magnitude & Phase

$$Z(n) = \underbrace{|Z(n)|}_{\text{magnitude}} \exp \left[\underbrace{j \arg\{Z(n)\}}_{\text{Phase}} \right]$$

Real & Imaginary parts

$$|Z(n)|^2 = \text{Re}^2\{Z(n)\} + \text{Im}^2\{Z(n)\}$$

Phase

$$\arg\{Z(n)\} = \tan^{-1} \frac{\text{Im}\{Z(n)\}}{\text{Re}\{Z(n)\}}$$

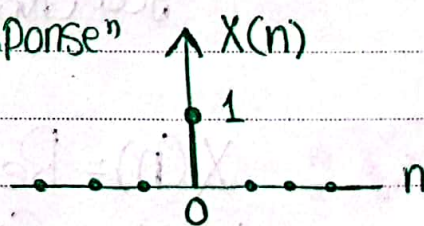
Complex Conjugate

$$Z^*(n) = \text{Re}\{Z(n)\} - j \text{Im}\{Z(n)\} = |Z(n)| \exp[-j \arg\{Z(n)\}]$$

$$\text{or } Z^*(n) = \text{Re}\{Z(n)\} - j \text{Im}\{Z(n)\} = |Z(n)| e^{-j\theta}$$

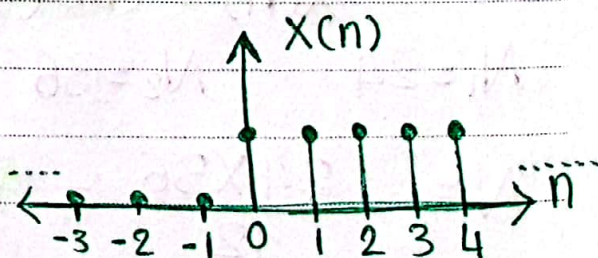
Unit Sample $\delta(n)$ "Impulse Response"

$$\delta(n) = \begin{cases} 1 & n=0 \\ 0 & \text{otherwise} \end{cases}$$



Unit Step $u(n)$

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$u(n) = \sum_{k=-\infty}^n \delta(k)$$

$$\delta(n) = u(n) - u(n-1) \quad \text{"midterm"}$$

exponential Sequence

$$X(n) = a^n$$

Complex exponential

$$e^{jn\omega_0} = \cos(n\omega_0) + j\sin(n\omega_0)$$

Periodic & Aperiodic Sequences

$$X(n) = X(n+N)$$

EX:

$$X_1(n) = a^n u(n) = \begin{cases} a^n & n \geq 0 \\ 0 & n < 0 \end{cases} \text{ "Aperiodic"}$$

$$X_2(n) = \cos(n^2) \text{ "Aperiodic" or "not periodic"}$$

$$X_3(n) = e^{j\frac{n\pi}{8}} \text{ "periodic" } \underline{N=16}$$

$$e^{jn\omega} = e^{j\frac{n\pi}{8}} \quad \omega = \frac{2\pi}{N} = \frac{\pi}{8} \quad N = \frac{2\pi \times 8}{\pi} = 16$$

Sum & Product of Two periodic Signals :

$$X(n) = X_1(n) + X_2(n)$$

$$N = \frac{N_1 N_2}{\text{gcd}(N_1, N_2)} \text{ "the greatest Common divisor" of } N_1 \text{ \& } N_2$$

EX:

$$X(n) = \text{Re}\{e^{j\frac{n\pi}{12}}\} + \text{Im}\{e^{j\frac{n\pi}{18}}\}$$

$$X(n) = \cos\left(\frac{n\pi}{12}\right) + \sin\left(\frac{n\pi}{18}\right)$$

$$N_1 = 24$$

$$N_2 = 36$$

$$N = \frac{24 \times 36}{12} = 72$$

"midterm"

$$EX_2: X(n) = e^{j\frac{\pi}{16}n} \cos\left(\frac{n\pi}{17}\right)$$

$$N_1 = 32$$

$$N_2 = 34$$

$$N = \frac{32 \times 34}{2} = 544$$

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Periodic Sequence

$$y(n) = \sum_{k=-\infty}^{\infty} x(n - kN)$$

N is +ve integer

Symmetric Sequences

Even $x(n) = x(-n)$

Odd $x(n) = -x(-n)$

$$x(n) = x_e(n) + x_o(n)$$

$$x_e(n) = \frac{1}{2} \{ x(n) + x(-n) \}$$

$$x_o(n) = \frac{1}{2} \{ x(n) - x(-n) \}$$

Complex Sequences

$$x(n) = x^*(-n) \quad \text{"Conjugate Symmetric"}$$

$$x(n) = -x^*(-n) \quad \text{"Conjugate antisymmetric"}$$

Ex:

$$x(n) = u(n)$$

"midterm"

$$x_e(n) = \frac{1}{2} [x(n) + x(-n)]$$

$$x_e(n) = \frac{1}{2} [u(n) + u(-n)]$$

even part:

$$x_e(n) = \frac{1}{2} + \frac{1}{2} \delta(n)$$

$$= \begin{cases} 1 & n=0 \\ \frac{1}{2} & n \neq 0 \end{cases}$$

$$x_o(n) = \frac{1}{2} [x(n) - x(-n)]$$

$$x_o(n) = \begin{cases} \frac{1}{2} & n > 0 \\ 0 & n = 0 \\ -\frac{1}{2} & n < 0 \end{cases}$$

Odd part: $x_o(n) = \frac{1}{2} \text{sgn}(n)$

$\text{sgn}(n)$ is the Signum Function. with ~~$x(n) = u(n)$~~

Subject: $X(n) = \alpha^n u(n)$

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The even part is:

$$X_e(n) = \frac{1}{2} [\alpha^n u(n) + \alpha^{-n} u(-n)] = \begin{cases} \frac{1}{2} \alpha^n & n > 0 \\ 1 & n = 0 \\ \frac{1}{2} \alpha^{-n} & n < 0 \end{cases}$$

$$X_e(n) = \frac{1}{2} \alpha^{|n|} + \frac{1}{2} \delta(n)$$

Odd part is:

$$X_o(n) = \frac{1}{2} [\alpha^n u(n) - \alpha^{-n} u(-n)] = \frac{1}{2} \alpha^{|n|} \text{sgn}(n)$$

Ex: $X(n) = \alpha^n u(n)$ ↗

Find Conjugate Symmetric part of $X(n)$.

$$X(n) = j e^{\frac{jn\pi}{4}}$$

or "midterm"

$$X_e(n) = \frac{1}{2} [X(n) + X^*(-n)] = \frac{1}{2} [j e^{\frac{jn\pi}{4}} - j e^{\frac{jn\pi}{4}}] = 0$$

this sequence is Conjugate antisymmetric

✱ Lec 2

Shifting

$$y(n) = x(n - n_0)$$

ex:

if n_0 is +ve number $\Rightarrow$ Shift Right $X(n-1)$

if n_0 is -ve number $\Rightarrow$ Shift Left $X(n+1)$

Reversal

$$y(n) = x(-n)$$

$$\uparrow (n - (-1))$$

Time Scaling

$$f(n) = Mn \quad (\text{down-sampling})$$

$$f(n) = \frac{n}{N} \quad (\text{up-sampling})$$

$$y(n) = \begin{cases} x\left(\frac{n}{N}\right) & n=0, \pm N, \pm 2N \\ 0 & \text{otherwise} \end{cases}$$

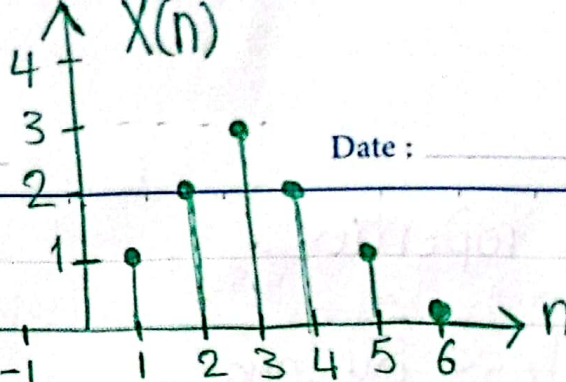
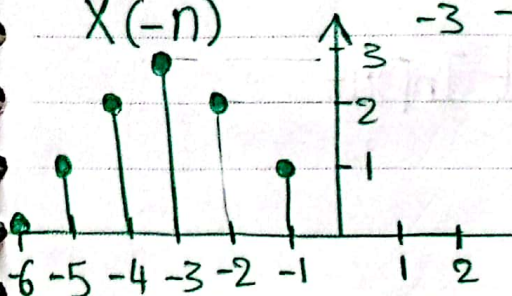
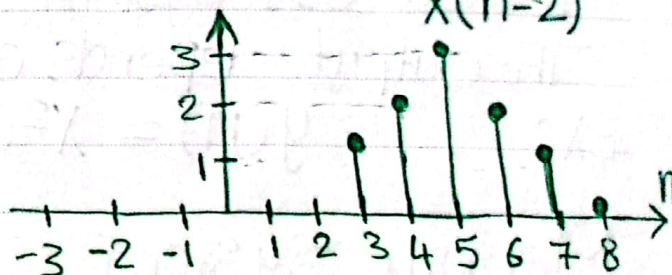
$$y(n) = x(f(n))$$

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Discrete-time Signal

 $X(-n)$  $X(n-2)$ 

Time Reversal

Shift to right by 2

X-axis $\Rightarrow 1+2=3, 2+2=4, 3+2=5, 4+2=6$

$$X(n) \xrightarrow[\text{or Delay}]{\text{Shifting}} X(n-n_0) \xrightarrow[\text{Reversal}]{\text{Time}} X(-n-n_0)$$

$$X(n) \xrightarrow[\text{Reversal}]{\text{Time}} X(-n) \xrightarrow[\text{or Delay}]{\text{Shifting}} X(-n+n_0)$$

Addition

$$y(n) = X_1(n) + X_2(n) \quad -\infty < n < \infty$$

Multiplication

$$y(n) = X_1(n) X_2(n) \quad -\infty < n < \infty$$

Scaling

$$y(n) = C X(n) \quad -\infty < n < \infty$$

Signal Decomposition

$$X(n) = \dots X(-1)\delta(n+1) + X(0)\delta(n) + X(1)\delta(n-1) + X(2)\delta(n-2) + \dots$$

$$X(n) = \sum_{k=-\infty}^{\infty} X(k)\delta(n-k)$$

Discrete Time Systems

$$X(n) \rightarrow \boxed{T[\cdot]} \rightarrow y(n) = T[X(n)]$$

System Properties

Memoryless System

The output depends on **Present Input**

Ex: $y(n) = x^2(n)$

Memory System

The output depends on Past or Future Input

Ex: $y(n) = x(n) + x(n-1)$

Past $\uparrow$ Value. $(n+1) \Rightarrow$ Future Value

Additivity

$$T[x_1(n) + x_2(n)] = T[x_1(n)] + T[x_2(n)]$$

Homogeneity

$$T[Cx(n)] = C T[x(n)]$$

Ex:

$$y(n) = \frac{x^2(n)}{x(n-1)}$$

mid term

$$y_1(n) =$$

$$T[x_1(n) + x_2(n)] = \frac{(x_1(n) + x_2(n))^2}{x_1(n-1) + x_2(n-1)}$$

$$y_2(n) =$$

$$T[x_1(n)] + T[x_2(n)] = \frac{x_1^2(n)}{x_1(n-1)} + \frac{x_2^2(n)}{x_2(n-1)}$$

$$y_1(n) \neq y_2(n)$$

“ **NOT Additive** ”

$$T[Cx(n)] = \frac{(Cx(n))^2}{Cx(n-1)} = C \frac{x^2(n)}{x(n-1)} = C T[x(n)]$$

“ **Homogeneous** ”

Ex: $y(n) = x(n) + x^*(n-1)$

$$y_1(n) = [X_1(n) + X_2(n)] + [X_1(n-1) + X_2(n-1)]^* =$$

$$y_2(n) = [X_1(n) + X_1^*(n-1)] + [X_2(n) + X_2^*(n-1)]$$

$$y_1(n) = y_2(n) \quad \text{« Additive »}$$

$$y_1 = T[CX(n)] = CX(n) + C^*X^*(n-1)$$

$$y_2 = C^T[X(n)] = CX(n) + C^*X^*(n-1)$$

« NOT Homogeneous »

Linear Systems

Homogeneous & Linear

$$T[a_1 X_1(n) + a_2 X_2(n)] = a_1 T[X_1(n)] + a_2 T[X_2(n)]$$

Superposition Summation

$$y(n) = T[X(n)] = T\left[\sum_{k=-\infty}^{\infty} X(k) \delta(n-k)\right] = \sum_{k=-\infty}^{\infty} T[X(k) \delta(n-k)]$$

$$y(n) = \sum_{k=-\infty}^{\infty} T[X(k) \delta(n-k)] = \sum_{k=-\infty}^{\infty} X(k) T[\delta(n-k)]$$

$$h_k(n) = T[\delta(n-k)]$$

$\Downarrow$
 $h_k(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} X(k) h_k(n) \quad \text{« Superposition Summation »}$$

Shift-Invariance

$$y(n-n_0) = T[X(n-n_0)]$$

Ex:

$$y(n) = X^2(n)$$

$$X(n) = X(n-n_0)$$

$$y_1(n) = [X(n)]^2 = X^2(n-n_0)$$

$$y_2(n-n_0) = X^2(n-n_0)$$

$$y_1(n) = y_2(n)$$

« Shift Invariance »

$$y(n) = x(n) + x(-n) \quad x(n) = \delta(n)$$

$$y(n) = \delta(n) + \delta(-n) = 2\delta(n)$$

$$y_1(n) = y(n-n_0) = \delta(n-n_0) + \delta(-n-n_0)$$

$$y_2(n) \neq 2\delta(n-n_0) \quad \text{"Shift-Varying"}$$

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$$y(n) = x(-n)$$

$$y_1(n) = x(-n-n_0)$$

$$y_2(n) = y(n-n_0) = x[-(n-n_0)] = x(-n+n_0)$$

$$y_1 \neq y_2 \quad \text{"Shift Varying"}$$

Linear Shift-Invariant Systems (LSI systems)

A system that is both Linear & Shift-Invariant is referred to as LSI system.

$h(n)$ is the response of an LSI system to the unit Sample $\delta(n)$.

$h(n-k)$ the response will be $\delta(n-k)$

$$h_k(n) = h(n-k)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

"Convolution Sum"

$$y(n) = x(n) * h(n)$$

Causality

output depends on **Present** or **Past** input

Non-Causal

output depends on **Future** Value.

EX:

$$y(n) = x(n) + x(n-1)$$

"Casual"

$$y(n) = x(n) + x(n+1)$$

"Non Casual"

Future

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midterm

Stability BIBO "bounded Input bounded output"

* For LSI System Stability is guaranteed if unit Sample response is Summable.

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

Ex: $h(n) = a^n u(n)$ $|a| < 1$ "midterm"

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=0}^{\infty} |a|^n = \frac{1}{1-|a|} \quad |a| < 1$$

"Stable"

الفكرة دي موجودة في الجدول (حفظ)

$$y(n) = n x(n)$$

$$x(n) = u(n) \quad y(n) = n u(n) \quad \text{"unbounded"}$$

"Not Stable"

Invertibility : input uniquely determined from the output.

Given any two inputs $x_1(n)$ & $x_2(n)$ with $x_1(n) \neq x_2(n)$ it must be true that $y_1(n) \neq y_2(n)$.

Ex: $y(n) = x(n) g(n)$

$$\text{if } g(n) \neq 0 \quad x(n) = \frac{y(n)}{g(n)} \quad \text{"Invertible"}$$

* $\nexists$ 2 diff. inputs بيطلقوا نفس output $\Rightarrow$ Not Invertible

Ex: $y(n) = 2x(n)$

$$x(n) = 0.5 y(n) \quad \text{"Invertible"}$$

« Convolution »

$$X(n) * h(n) = \sum_{k=-\infty}^{\infty} X(k) h(n-k)$$

Properties

Commutative Property

$$X(n) * h(n) = h(n) * X(n) \text{ «order isn't important»}$$

Associative Property

$$\{X(n) * h_1(n)\} * h_2(n) = X(n) * \{h_1(n) * h_2(n)\}$$

Distributive Property

$$X(n) * \{h_1(n) + h_2(n)\} = X(n) * h_1(n) + X(n) * h_2(n)$$

Ex: $X(n) = a^n u(n) = \begin{cases} a^n & n \geq 0 \\ 0 & n < 0 \end{cases}$

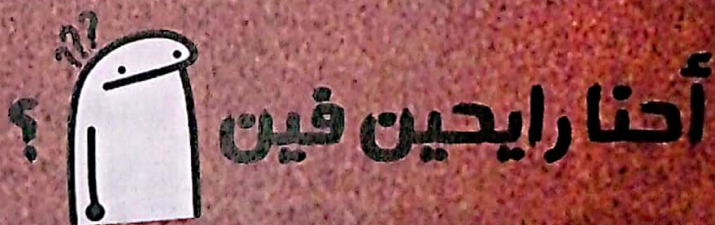
$$h(n) = u(n)$$

$$y(n) = X(n) * h(n) = \sum_{k=-\infty}^{\infty} X(k) h(n-k) =$$

$$\sum_{k=-\infty}^{\infty} a^k u(k) u(n-k)$$

$$y(n) = \sum_{k=0}^n a^k = \frac{1-a^{n+1}}{1-a}$$

$$y(n) = \frac{1-a^{n+1}}{1-a} u(n)$$



Subject:

RULES

Date:

Table 1.1

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$$

Another Form $\Rightarrow$

$$\sum_{n=0}^N a^n = \frac{1-a^{N+1}}{1-a}$$

$$\sum_{n=0}^{N-1} na^n = \frac{(N-1)a^{N+1} - Na^N + a}{(1-a)^2}$$

$$\sum_{n=0}^{N-1} n = \frac{1}{2}N(N-1)$$

مبتسم رغم الإعاقة

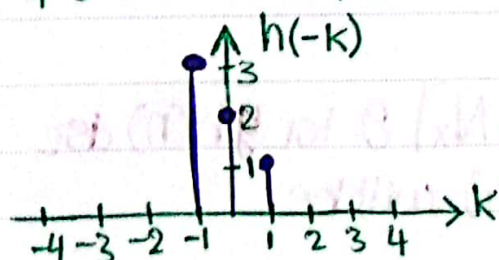
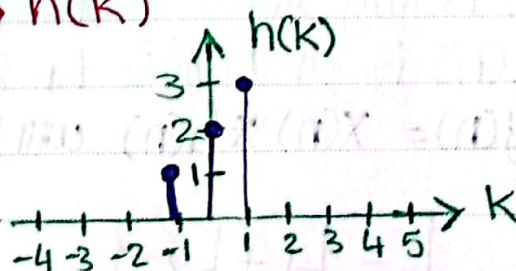
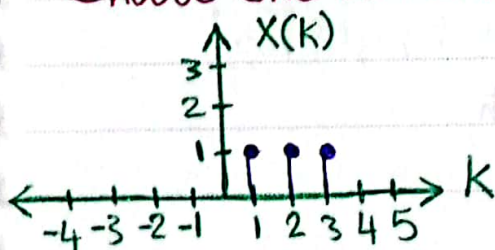
$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad |a| < 1$$

$$\sum_{n=0}^{\infty} na^n = \frac{a}{(1-a)^2} \quad |a| < 1$$

$$\sum_{n=0}^{N-1} n^2 = \frac{1}{6}N(N-1)(2N-1)$$

Ex: $y(n) = x(n) * h(n)$

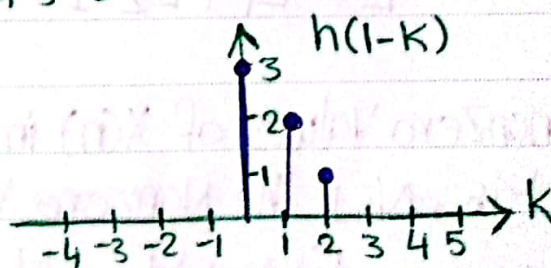
$x(k)$ & $h(k)$ are both plotted as a function of k . So we choose one to reverse in time $\rightarrow h(k)$



Note

 $h(1-k)$ $h-(k-1)$

Shift to Right



"Time Reverse"

 $y(n) = x(n-n_0)$

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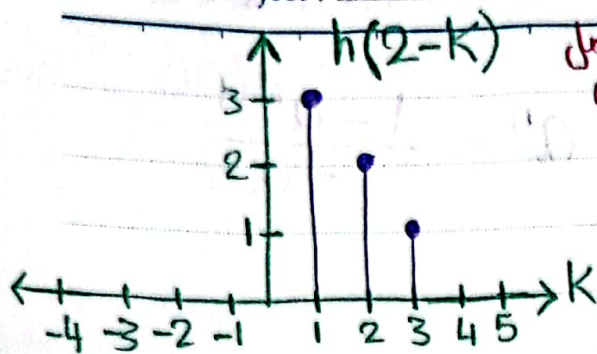
"Shifting"

$h-(k-2)$
"Shift to the right by 2"

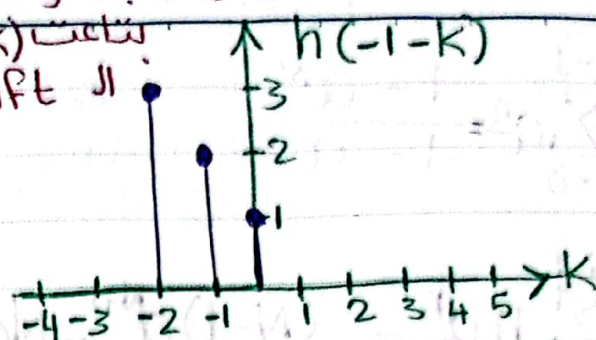
$h-(k+1)$
"Shift to the Left by 1"

Subject :

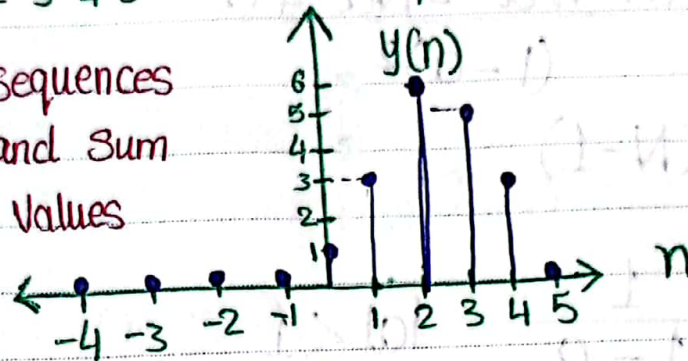
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$h(k)$ بتغير
"Shift"



Multiply the two sequences
 $X(k)$ & $h(n-k)$ and Sum
the product for all values
of k . The result
will be $y(n)$



$$y(0) = 1, y(1) = 3, y(2) = 6, y(3) = 5, y(4) = 3$$

$$y(n) = 0 \quad n > 4, \quad y(-1) = 0, \quad y(n) = 0 \quad n < 0$$

$$* y(2) = \sum_{-\infty}^{\infty} x(k)h(2-k) = 6$$

$$* y(1) = \sum_{-\infty}^{\infty} x(k)h(1-k) = 3$$

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Performing Convolution of two Finite-length sequences
if $x(n)$ is of Length L_1 & $h(n)$ is of length L_2
 $y(n) = x(n) * h(n)$ will be of Length

$$L = L_1 + L_2 - 1$$

if nonzero values of $x(n)$ interval $[M_x, N_x]$ & for $h(n)$ are
 $[M_h, N_h]$ the Nonzero values of $y(n)$ will be
 $[M_x + M_h, N_x + N_h]$

$$x(k) \Rightarrow [1, 3], \quad h(k) = [-1, 1] \quad y(n) = [0, 4]$$

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Ex:

$$X(n) = \begin{cases} 1 & 10 \leq n \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

$$h(n) = \begin{cases} n & -5 \leq n \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$X(n) = \underset{M_x}{[10, 20]} \quad h(n) = \underset{M_h}{[-5, 5]} \quad y(n) = [5, 25]$$

$$y(n) = [M_x + M_h, N_x + N_h]$$

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